

East Brandenburg

Full data report

Generated 2026-08-21

Brandenburg

NUTS-3 regions: DE409



Figure 1: The East Brandenburg Living Lab: study region (left) and its location within Germany (right). Basemap: Tiles © Esri - Source: Esri, i-cubed, USDA, USGS, AEX, GeoEye, Getmapping, Aerogrid, IGN, IGP, UPR-EGP, and the GIS User Community

Agriculture

Key figures

CROPLAND AREA	NUMBER OF FARMS	AVERAGE FARM SIZE	SHARE OF ORGANIC FARMING
252,996 ha	1,374 count	220.7 ha	12.4 %

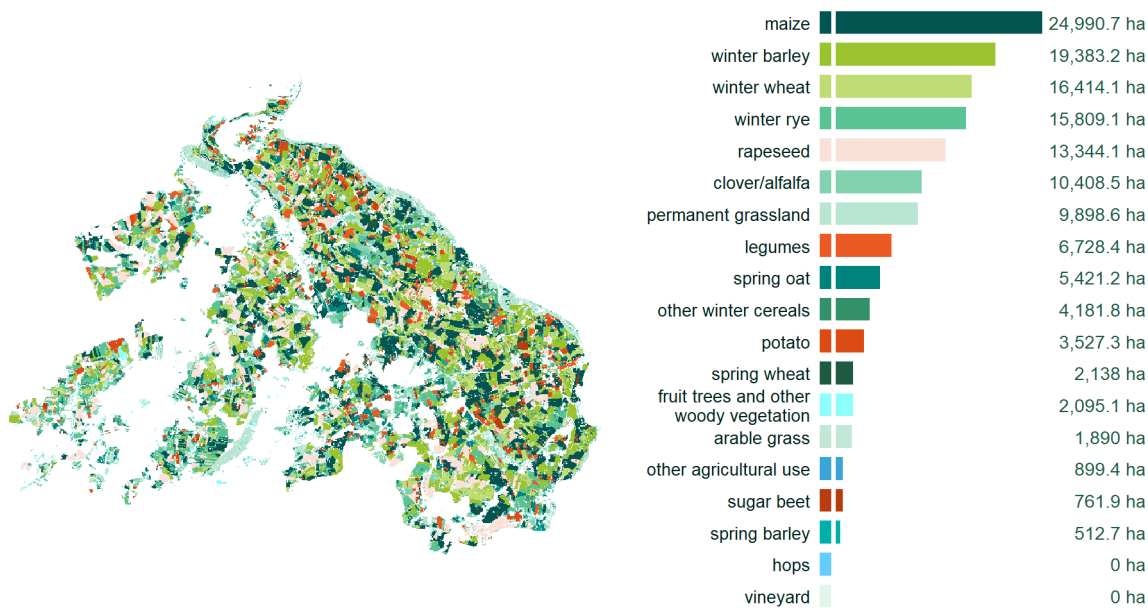


Figure 2: Map showing Agriculture in the East Brandenburg Living Lab (Data: Crop types (DLR, 2024))

Overview

Main crops: winter grain, maize for silage/energy, winter rapeseed, Other land uses: horticulture and seed/plant cultivation minimal; permanent grassland. Animal husbandry: poultry most common; pigs, cattle, sheep, goats declining.

Challenges

Greater irrigation demand and competition for scarce water resources. Reduced crop productivity and stability under intensifying climate stress. High adaptation pressure on farmers, requiring new practices and investments. Competition for land between agriculture and renewable energy expansion. Decline in livestock diversity (pigs, cattle, sheep, goats). Limited horticulture and specialized crop production (low diversification). Balancing organic and conventional farming practices while maintaining productivity.

Climate

Key figures

GROWING DEGREE DAYS, ANNUAL SUM (BASE 5 °C)	MEAN ANNUAL TEMPERATURE	ANNUAL PRECIPITATION	PRECIPITATION OF WARMEST QUARTER
2,011 °C·d +1,135 °C·d by 2071-2100	9.3 °C +4.1 °C by 2071-2100	573 mm +3 % by 2071-2100	186 mm -5.1 % by 2071-2100



Figure 3: Map showing Climate in the East Brandenburg Living Lab (Data: Climate (CHELSA V2.1 / CMIP6): Growing degree days, Mean annual temperature, Annual precipitation, Precipitation of warmest quarter)

Heat accumulated above 5 °C over the year - a measure of how much growing season a crop gets.

Mean annual air temperature, averaged over the reference period.

Total precipitation summed over the year.

Precipitation falling in the warmest three-month quarter - relevant to summer drought stress.

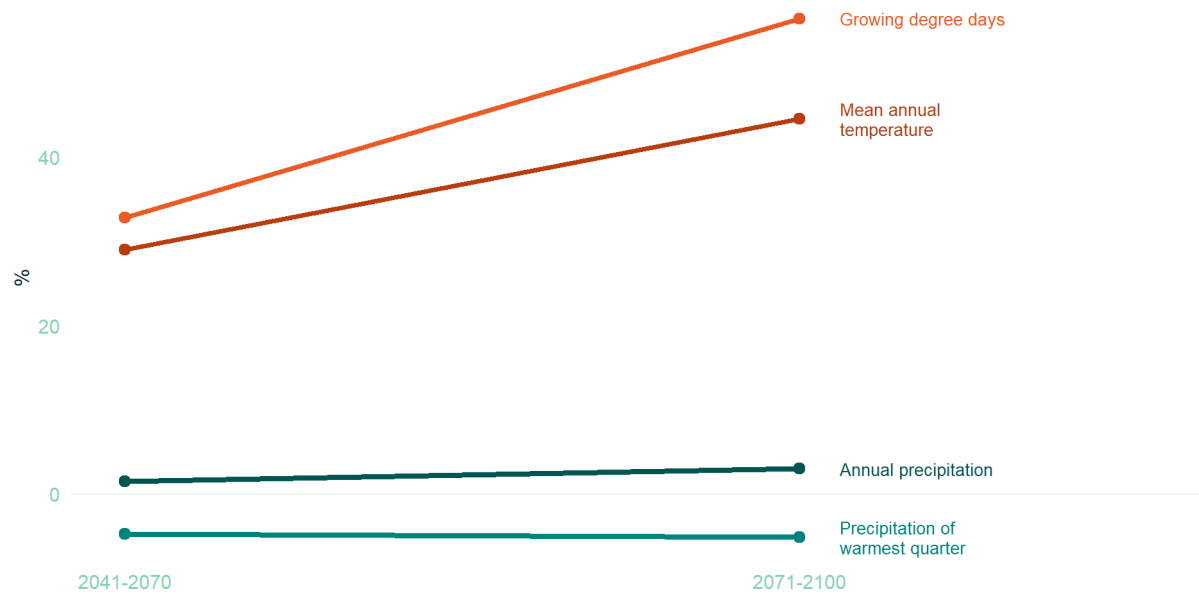


Figure 4: Chart showing Climate in the East Brandenburg Living Lab (Data: Climate (CHELSA V2.1 / CMIP6): Growing degree days, Mean annual temperature, Annual precipitation, Precipitation of warmest quarter)

Overview

Continuous warming trend observed since 1881: +1.3 °C increase in mean annual temperature to date. Short-term climate projections (2021—2050): Additional warming of +1.1 to +1.5 °C. Shift in climate extremes: More hot summer days, Fewer frost days, hail and storm, spring drought increases expected in eastern Brandenburg due to stronger continental influence

Challenges

High drought vulnerability due to limited soil water retention capability

Soil

Key figures

NITROGEN SURPLUS	PHOSPHORUS SURPLUS	GROUNDWATER ABSTRACTION (NON-PUBLIC SUPPLY)
No data available kg N/ha UAA	No data available kg P/ha UAA	20,656 1000 m3

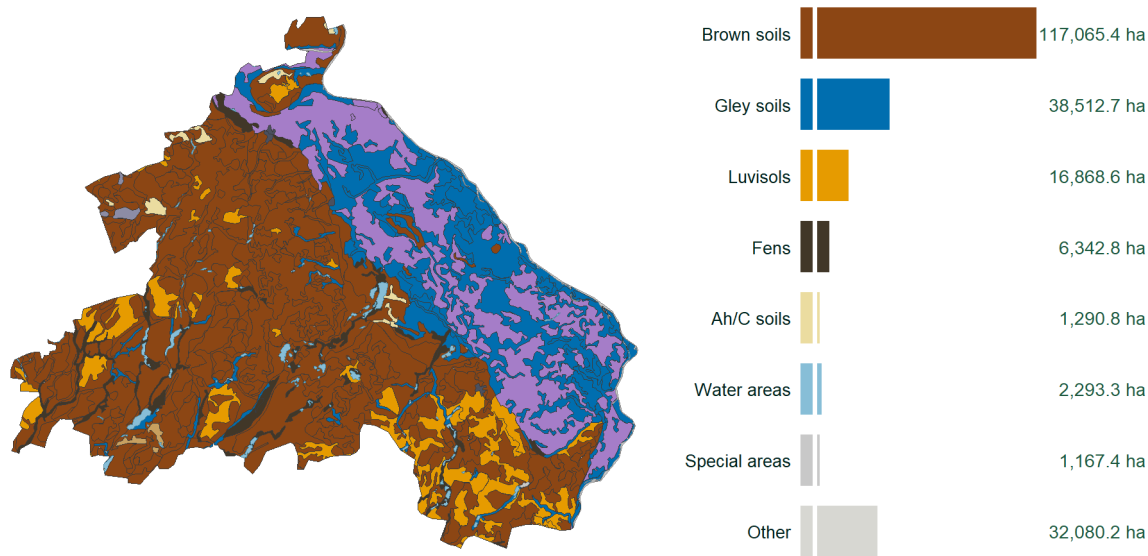


Figure 5: Map showing Soil in the East Brandenburg Living Lab (Data: Soil overview map (BUEK250))

Legend shows the dominant semantic soil groups for this Living Lab; raw BUEK IDs stay in the data only as provenance.

Overview

Dominated by sandy and loamy-sandy soils ('Märkischer Sand') with low natural fertility. Müncheberger Soil Rating: Mostly very low to low soil quality (below 50 points); medium-quality soils only in specific floodplain regions (e.g., Oderbruch). High spatial heterogeneity of soil quality within single fields due to glacial landscape formation

Socio-economic

Key figures

TOTAL POPULATION	GDP PER CAPITA	UNEMPLOYMENT RATE	DISPOSABLE HOUSEHOLD INCOME PER CAPITA
704,008 persons	33,837.25 EUR	6.25 %	26,526.75 EUR

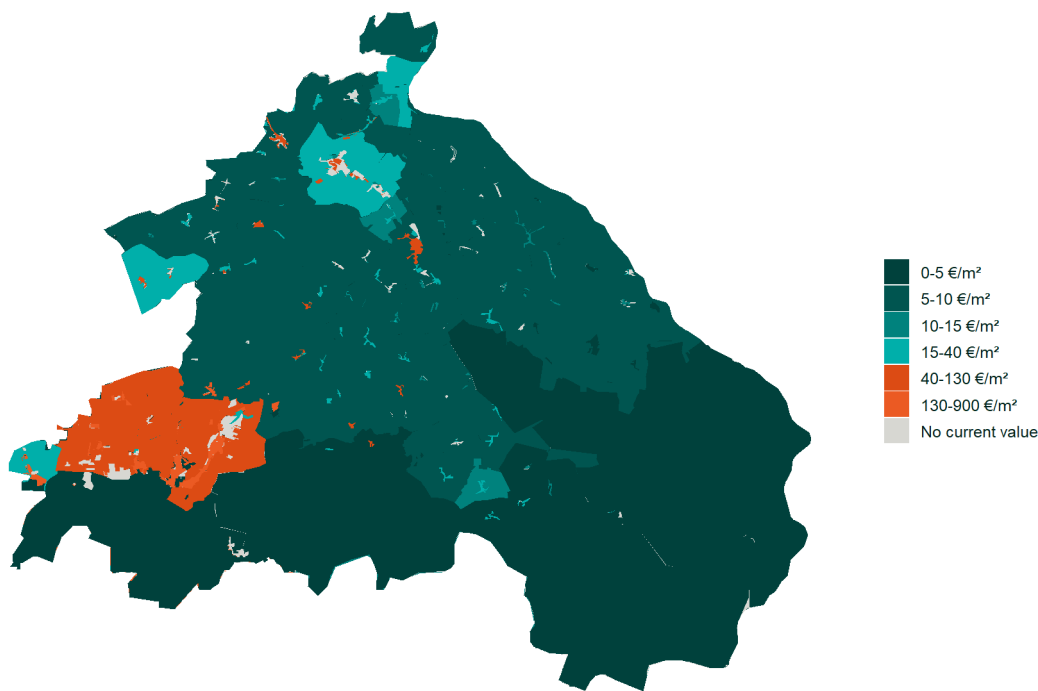


Figure 6: Map showing Socio-economic in the East Brandenburg Living Lab (Data: Standard land values (BORIS))

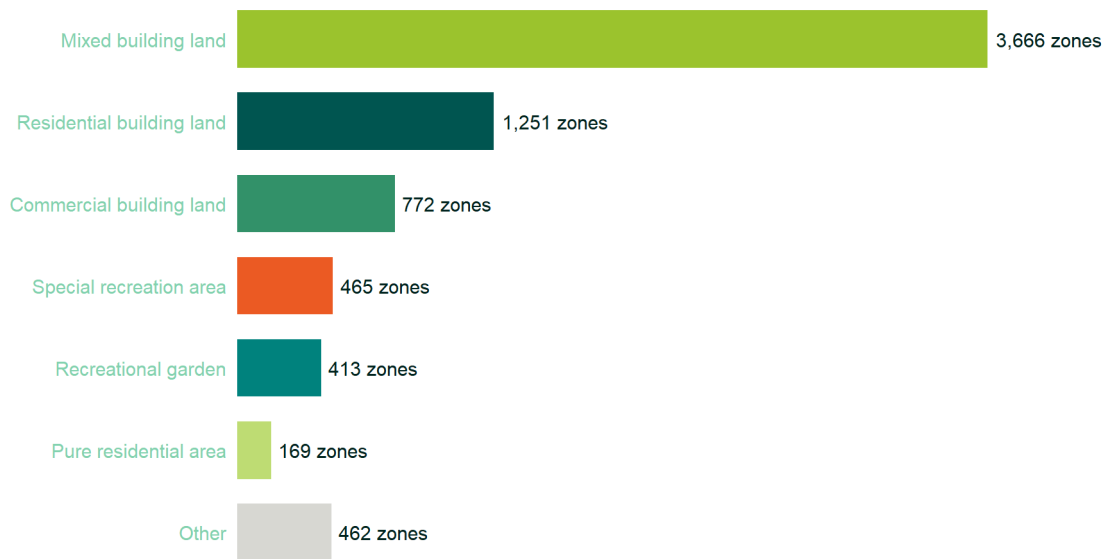


Figure 7: Chart showing Socio-economic in the East Brandenburg Living Lab (Data: Standard land values (BORIS))

Overview

Higher population density and employment near Berlin; commuter flows mainly from eastern border toward Berlin. Low population density, aging population, and outmigration of young people reduce workforce availability. Employment structure: Predominantly services (tourism, healthcare, social services), followed by manufacturing/trades; agriculture employs only a small number of people despite large land areas. Economic hubs: Tesla Gigafactory and Berlin-Brandenburg Airport (LOS) attract businesses; Berlin metro area provides market opportunities. Growth trends: Commercial development and employment expected to increase. Tourism: Driven by nature parks, protected areas, and spa towns.

Challenges

Agriculture and peripheral industries generate limited regional value creation. Weak local processing and value-adding facilities limits regional economic integration, many areas faced a decline of traditional manufacturing and industrial sites. Skilled labor shortage (Fachkräftemangel) compounded by competition with Berlin's dynamic job market

Landscape

Key figures

FOREST AREA	NATURA 2000 AREA	NATURE RESERVES AREA	SHARE OF SEALED SURFACE
278,900 ha	53,054.8 ha	9,656.9 ha	11.4 %

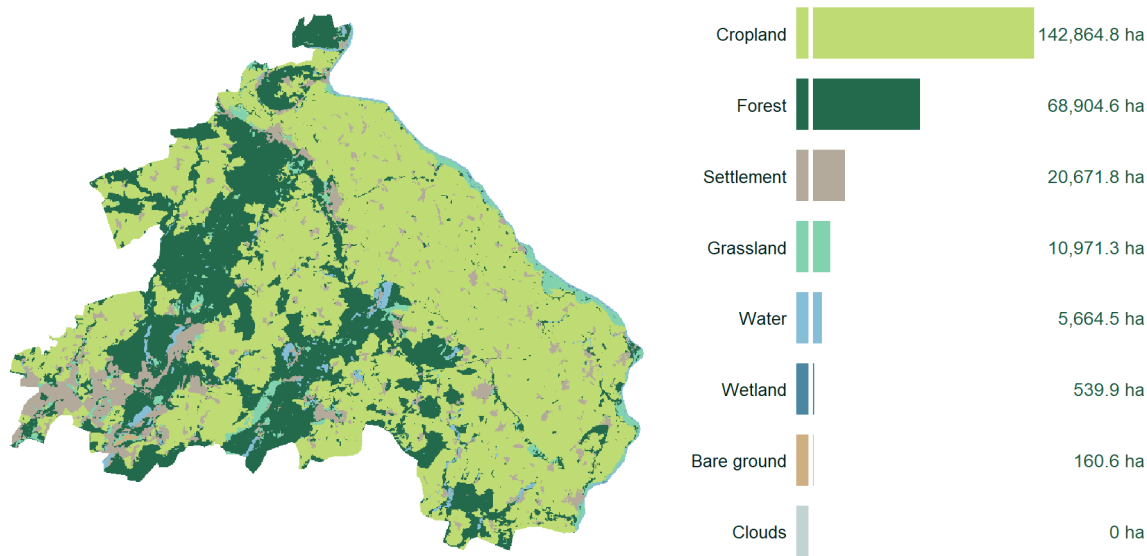


Figure 8: Map showing Landscape in the East Brandenburg Living Lab (Data: Land cover (Esri/Impact Observatory, 2024))

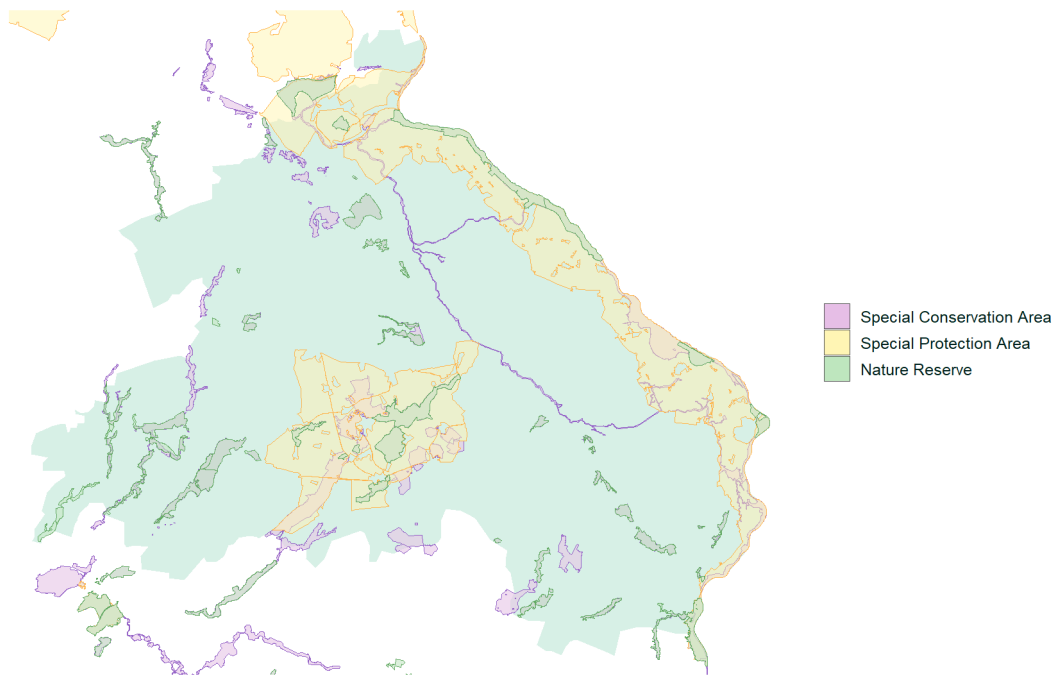


Figure 9: Map showing Protected Areas in the East Brandenburg Living Lab (Data: Protected areas (BfN, 2019/2023))

Conservation sites intersecting this Living Lab region. Full polygon boundaries are shown, so sites extend beyond the region outline and designations may overlap. Source: BfN (FFH and bird sanctuaries 2019, nature reserves 2023) - not suitable for planning purposes.

Overview

Lowland region shaped by multiple ice age glaciations. Minimal elevation differences with predominantly flat terrain with Oderbruch as a wide river valley shaped by glacial meltwater

Challenges

Biodiversity decline in agricultural landscapes (e.g., drying of breeding habitats for amphibians). Land ownership: much farmland not owned by farmers.